

Recognition and reconstruction of tri-dimensional objects within urban meshes

Rodrigo Garcia¹, Mauro Farracha¹, José Dionísio², Alexandre Bernardino³, João Paulo Costeira³

ISR/DEEC^{1,3}, IDMEC-Polo Lisboa²

Instituto Superior Técnico

Av. Rovisco Pais, 1049-001 Lisboa

{rjsdg,mdfa}@mega.ist.utl.pt¹, jdionisio@ist.utl.pt², {alex,jpc}@isr.ist.utl.pt³

Resume

The present paper describes a set of techniques in the field of recognition and reconstruction of 3D objects from geo-referenced data. The available data set is a structured grid based on airborne-laser measures over the city of Lisbon. The main focus of this work is on the object detection phase using the DSM (Digital surface Model). Based on that recognition, the building entities are extracted upon a set of characteristic parameters, like height gradient, among others. Distinct approaches and variants are described towards a geometric reconstruction for future real time edition and visualization within interactive virtual environments.

Keywords

Laser airborne data, DSM, Region Growing, Best Fit Ellipse, 3D Reconstruction, Interactive Virtual Environments, Real Time Visualization

1. INTRODUCTION

The high pressure on urban planners has stimulated a great deal of research in the area of automatic and semi-automatic tools for acquisition and validation of cadastral data. Among others, Urbanism departments have two main interests in these tools: knowing what really exists in a certain urban space and how would it look like if changed in a certain way (e.g. after inserting a new building) [9].

An increasing number of geoinformation users require more and more automated systems for 3D city model building. Due to the advantage of being an active technique, airborne laser scanning is becoming a key source of information for complex altimetric model creation. Available data, both photometric and range, allowed the validation of several feature grouping algorithms for 3D model approximation.

In this paper we describe a methodology that, by processing airborne laser data, creates a virtual 3D scene of the environment. Furthermore, users are able to interactively edit the scene, and navigate through a photorealistic environment. in real time.

2. PROBLEM DEFINITION

Given a huge set of airborne-based laser data, the main problem to addresses is finding best fitting algorithms for the data handling pipeline: data evaluation, pattern identi-

fication, segmentation and visualization towards an interactive scenario that represents 3-dimensionally the altimetric data set. Moreover, once available, the framework should include support for the comparison with former versions towards automatic update.

3. RELATED WORK

Given the main topic of the research, a special emphasis was given to recognition and object reconstruction based on altimetric data. The available literature, beyond computer graphics, lead us to two distinct approaches depending mainly on the data itself, that means, depending on the availability of the DTM (Digital Terrain Model) or the DSM (Digital Surface Model).

First of all, DSM represents in this text, the original raw airborne-based data set, with a co-ordinate Z for each pair of (x,y) values. (Fig. 1). Brighter areas stand for higher Z co-ordinates. On the other side, DTM represents a set of this data, since it stands only for the terrain. Buildings or vegetation, among other objects, are not included.

The work of Ansgar Brunn e Uwe Weidner 1998 [8] was the starting point for a data-handling concept. Their recent and relevant work has been often referenced and states a coherent approach following the basic principles of DSM and DTM use.



Fig. 1: grey-scale example of a DSM

Basically, they describe their research in three conventional phases, as follows:

- Detection of pattern characteristics (DSM)
- Domain segmentation (DSM)
- 2D and 3D reconstruction

although each one contains special variants which is worth to describe.

Firstly, they generate the normalized DSM, basically the difference between DTM e DSM, of the working area. If one imagines a couple of buildings on a valley as well as on the top of a mountain, its normalized DSM should look as a representation of such buildings on a common plane. This process makes easier the next step, the segmentation.

The same research group developed an approach based on a binary segmentation of the height fields. This way, given a threshold, it was possible to divide the initial set in two classes of elements (Fig. 2).



Fig. 2: Results of the DSM segmentation of Fig. 1

On one hand, we have the candidates for ground elements. On the other hand, we have the reminiscent instances. Although this classification brings a step forward in the recognition phase, since terrain recognition could allow an effective separation between data layers, the problem is far from being solved. In fact, the second

class of elements includes amalgams of data types: buildings, vegetation, and street utilities, among others. Even if one concentrates himself on two main data sets, buildings and vegetation, different algorithms must be used for a best-fit evaluation. One way to go deals with the computation of the surface rugosity by means of the normal variance at the surface, the local curvature degree to a specific point and the gradients (Fig. 3). The detected areas are consequently isolated and they are removed from the normalized DSM data set (Fig. 4).



Fig. 3: Segmentation after the computation of rugosities of the DSM



Fig. 4: Visual aspect of the differences map

This is an iterative process and usually conducts to fairly good results. However, the demand to a fixed threshold in each iteration falls on a type of problem that depends on the quality of the data set and on the operator's experience. Therefore, the authors suggest a way to minimize such constraints: the use of Baesian networks.

Baesian networks do not segmentate through a fixed threshold but through the use of classes of properties, referenced in this work as characteristics.

For each point of the data set one compute its probability of belonging to a building, for instance, according to the value of the characteristic in analysis. There is no limit to

the number of characteristics evolved in this process. Moreover, the user defines each characteristic and there are no predefined rules for appliance. That means, a characteristic may be the point's height, the local curvature, the gradient of the point, or other concepts that allow suitable characterization of points.

Based on these principles, follows the set up of the network and the computation of points which probability can be grouped by similarity. Although it seems an expedite method, the computation time is huge for the amount of data points usually used in such analysis.

Finally, the third conventional phase: after the detection of all segments that describe existing (or possible existent buildings) follows the development of the 3D reconstruction algorithms. For that, the same team uses prismatic and parametric mathematical models for tri-dimensional representation. Those models are built based on the analysis of the detected candidate segments. The estimation of the parameters does not include the roofs. In fact, the methods addresses exclusively on the estimation of planar surfaces contained in the normalized DSM. However, a further level of detail could bring to a problem of a multiple intersection of planar surfaces, which can lead on a rough roof definition. In fact, the entity called roof can be a hard problem to solve. This type of detail is also described in other contributions, which complement the work of *Brunn e Weidner* 1998 [8].

Steinle e Vögtle [7], for example, described a real case study of an earthquake and its impact on the structural changes of an urban model. In such scenario, two models were available, before the earthquake and after. The models were compared and the buildings with major probability of ruin were identified. To have their job done, they proposed five distinct analyses for object recognition: spectral signal, height above the floor, texture's height, normal vectors and object's shape.

Spectral signal – this is the only method which do not use the DSM and is recognized by this research group as the best method.

Height above the floor – similar to the one presented and described by *Brunn e Weidner* [8].

Texture's height - Uses the same definitions of the work described by *Maas* [1][2][3]. They are enough precise to separate buildings from vegetation and isolate the last ones.

Normal vectors – it consists on the detection of homogeneous areas in the DSM, that is, the detection of a planar surface given a set of normal vectors with a similar direction, or the detection of a candidate to vegetation, if the set of normal vectors have almost random directions.

Object's shape [4][5]- First of all, a set of selected points is connected by a triangular grid. Afterwards, an estimation of the shape of the building is performed. The height of the building is estimated, candidates to planar surfaces for the roof are selected

and, after the demanded intersections a first 3D model is obtained.

4. IMPLEMENTATION

As previously stated, many related works split the processing of the altimetry data set in the following phases: (i) computation of the DSM from raw 3D data; (ii) segmentation of relevant structures in the DSM (buildings, roads, ground, trees, parked cars) and; (iii) the definition of geometric 3D structures identified in the segmented DSM. In this work we follow a similar approach but present some novelties in the employed techniques. In particular, we propose:

- a binary DSM segmentation algorithm based on “region growing”, that allows non-uniform terrain elevation, thus avoiding the computation of the DTM;
- a boundary estimation algorithm based on “best fit ellipses”, to derive line segments from edges in the segmented maps, with computational improvements with regard to state-of-the-art techniques;
- a vegetation detection method based on entropy and variance of the surface normal vectors.

The above techniques are described in detail in the remaining of this section, with applications to a practical case.

4.1 Case study: downtown Lisbon

The Digital Surface Model (DSM) of Lisbon downtown is the starting point for the work presented in this paper (Fig. 5). We will describe the methodology employed to segment the buildings contained in the area, and reconstruct a 3D scene with planar surfaces, corresponding to buildings' boundaries.



Fig. 5: Partial DSM of Lisbon downtown (bright points indicate higher elevations).

The visual inspection of the DSM as in Fig. 5 does not reveal the noise and irregularities present in the representation. However, displaying the data as a surface (Fig. 6), shows that, in certain regions, the segmentation of distinct buildings is not obvious even for a human subject. Thus, both the region segmentation and the 3D reconstruction algorithms must take into account the “noisy” nature of the acquired data.

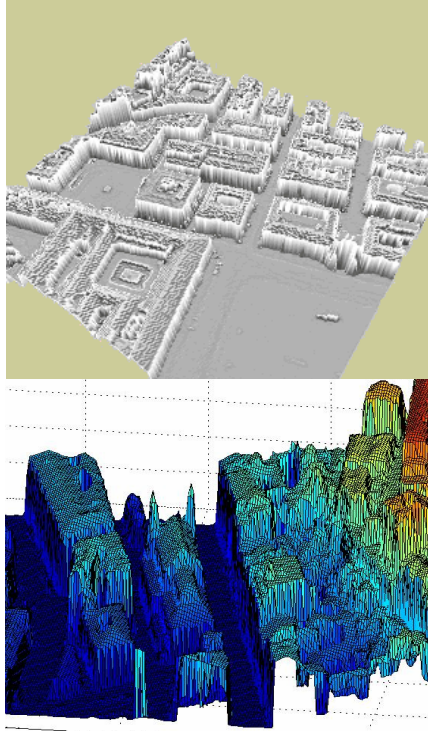


Fig. 6: Representation of downtown Lisbon in terms of surface map. In certain parts of the town, structures are very regular (top), but other regions are noisy and irregular (bottom).

The work described in this paper is schematically represented in the following diagram (Fig. 7):

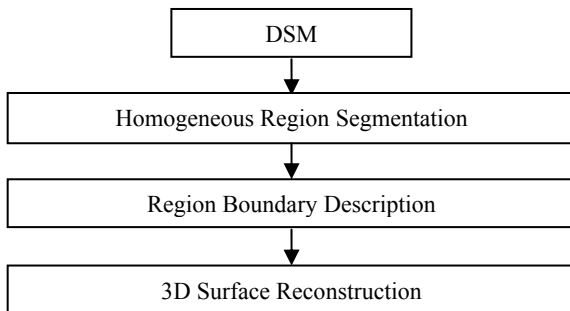


Fig. 7: Sequential steps of DSM processing.

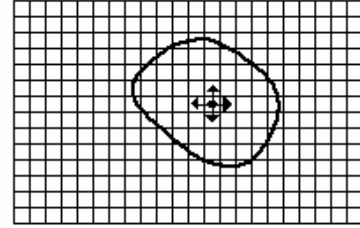
4.1.1 Homogeneous Region Segmentation

The first step of the processing sequence consists in detecting regions in the DSM having homogeneous features. We have used features like elevation (raw DSM data), elevation gradient, and surface normal vectors, among others.

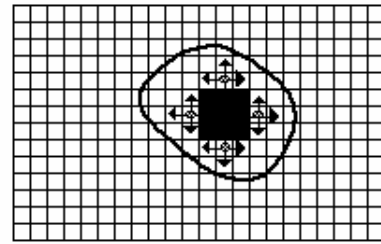
The segmentation algorithm is based on the “region growing” technique. Starting on a certain point (seed) in the DSM, the algorithm checks its neighbors and agglomerates the ones similar in terms of the considered

feature. This is a “bottom-up” approach, and finishes when all points in the DSM have been attributed to a region. The method is illustrated in Fig. 8, representing a region not yet segmented. The dark boundary line represents the region to segment, where the inner pixels have similar values of, for example, surface height. From the initial starting point, the algorithm sequentially compares the points in its neighborhood. If a neighbor point has a similar value of the feature, it is included to the region to which the seed point belongs. The arrows indicate the growing direction:

1) Initialization



2) Intermediate step



3) Final result

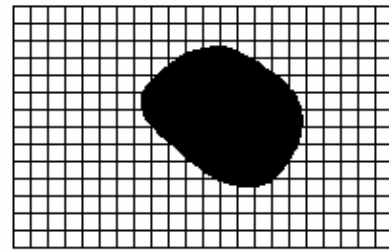


Fig. 8: Illustration of the “Region Growing” algorithm steps. The dots indicate the pixels being analyzed and the arrows show the growing direction. Dark pixels have been aggregated to the region.

In the end of the procedure we obtain regions that are considered homogeneous and represent the main objects in the scene. A Fig. 9 shown the result of the height based segmentation algorithm applied to the Lisbon downtown DSM. All connected white regions have been identified as non-ground objects.



Fig. 9: Results of the height-based “region growing” algorithm, applied to Lisbon downtown DSM.

The “Region Growing” algorithm detects the ground as a particular region. However it is easily discriminated from other objects because it has a lower elevation. Despite the same color (white) is used for the non-ground objects, the algorithm guarantees that non-connected objects are well segmented and individuated. Segmented objects that do not correspond to buildings will be rejected. This will be addressed latter in this section.

4.1.2 Region Boundary Description

Having identified regions corresponding to buildings in the DSM, the following step consists in identifying the region boundaries and line segments corresponding to the limits of the buildings. Then, we will derive the parameters corresponding to the location of buildings and their limits, their height, the shape of the roof, etc. These parameters will be used in the 3D-reconstruction step.

The boundaries of the regions are identified using edge detection operators on the segmented maps. The result of this operation is shown in Fig. 10.



Fig. 10: An example of boundary extraction in homogeneous region maps.

The edge operator computes a discrete approximation to the gradient of the segmented maps and selects the maxima of the gradient. Since the segmented maps are binary, the discontinuity detection is much more robust than the application of the edge operator to the raw DSM.

Once boundary points have been obtained, we are interested in identifying line segments constituting the external walls of the building. Then, the height of the walls can be obtained by picking the DSM values corresponding to the endings of the line segment.

The solution adopted to perform the identification of line segments is based in the “Best Fit Ellipse” algorithm. The algorithm consists in adjusting an ellipse to a certain set of data points, by the minimization of the squared approximation error. We describe a line segment as an ellipse with a large eccentricity (ratio between the larger and smaller axes). Given three points, we can adjust only one ellipse having the major axis delimited by two of those points. Let us consider a set of three points, composed by the extreme points (x_1, y_1) , (x_2, y_2) and an additional arbitrary point (x, y) (Fig. 11).

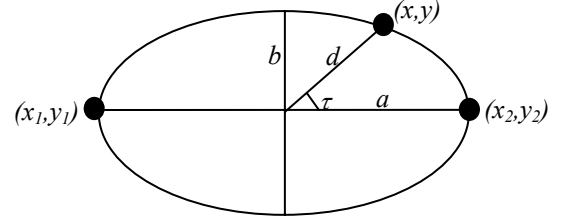


Fig. 11: Representation of an ellipse for the “Best Fit Ellipse” algorithm. a – major axis; b – minor axis; d – distance from a point to the center of the ellipse; τ – angle from a point to the major axis.

The major axis of the ellipse is given by half the distance between the extreme points:

$$a = \frac{\sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2}}{2}$$

Computing the distance d between point (x, y) and the center of the ellipse, and the angle τ made with the major axis, the minor axis b can be computed by:

$$b = \sqrt{\frac{a^2 d^2 \sin^2(\tau)}{a^2 - d^2 \cos^2(\tau)}}$$

Given a set of neighbor points in the boundary of a building, the algorithm computes the extreme points (points at maximal distance), and uses them to define the major axis. Then for each non-extreme point, it computes the minor axis of the ellipse fitting that point and the extreme ones. If, for a certain point in the set the ellipse eccentricity is smaller than a threshold (selected as 2, in our case), then this point should represent a line break, probably corresponding to a corner and the start of a new wall. The algorithm proceeds until all the boundary points can be represented by ellipses with large eccentricities.

When a line segment representing a wall is detected, we also compute its average height, width and direction, to be used in the 3D-reconstruction step.

An example, the results provided by this algorithm are shown in Fig. 12.

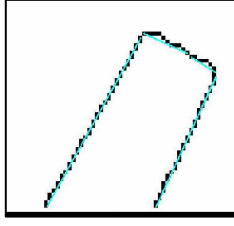


Fig. 12: Line segment estimation using the “Best Fit Ellipse” algorithm.

4.1.3 Tree Detection

In urban areas, most large structures correspond to human made buildings. However, regions with trees and bushes may be undesirably segmented by height based region growing algorithms. A characteristic feature that distinguishes these structures from buildings is that they can not be represented by a small set of planar surfaces, thus presenting large variations in surface orientations. To detect, early in the segmentation phase, regions of this type, we compute features based on the variance (Fig. 13) and entropy (Fig. 14) of the surface normal vectors in the DSM surface. Then, regions with high variance and entropy are excluded from potential segmentations.

Both the variance and the entropy are computed by the analysis of the vertical component of the surface normal vectors (with unit magnitude), on a region around the point of interest ($N \times N$ window). The variance is computed by:

$$V = \sum_i (Z_i - \bar{Z})^2$$

where Z_i are the vertical components of the surface and $\bar{Z}$ is their average in the computation window.



Fig. 13: Representation of part of the DSM, with high variance areas identified in black.

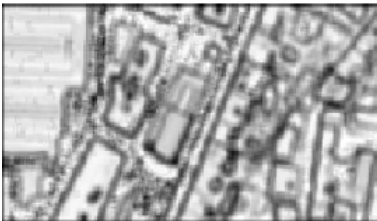


Fig. 14: Entropy map – high entropy values are represented in dark.

To compute the entropy, we first compute the histogram of vertical normal component in the analysis window, $P(h_i)$, where h_i represent the histogram bins and P is the relative frequency of occurrence of the surface normal vectors. The entropy of a region is, then, computed by:

$$E = \sum_i -P(h_i) \times \log(P(h_i))$$

Finally, points of high entropy or variance can be removed from the DSM to avoid segmenting undesired objects.

4.1.4 3D Reconstruction

The last step of the processing sequence is the representation of the identified building parameters in a 3D-object hierarchy. The purpose of this step is the real-time visualization and navigation in a virtual environment. A diagram of the adopted object hierarchy is shown in Fig.15. Obviously the hierarchy can be extended. In future work we intend to agglomerate buildings into quarters, neighborhoods, etc.

Right now the 3D representation is composed by buildings constituted by several connected planar surfaces that correspond to the external wall and roof. Results are shown in Fig. 16.

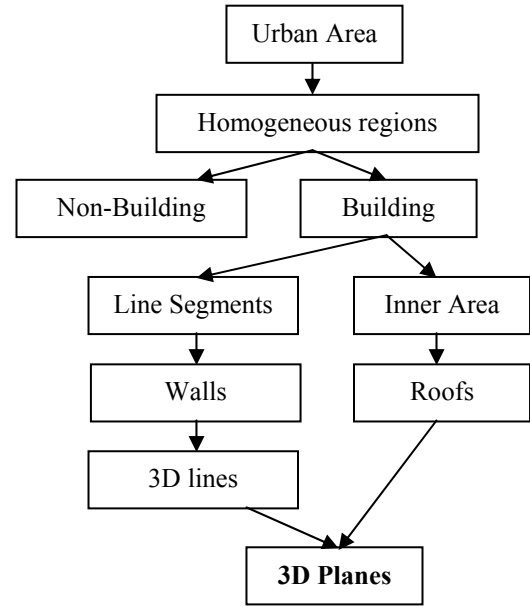


Fig 15. Object hierarchy for 3D representation.

